Proof of Concept: Use and Fine-Tune MACE

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Outline

Examples of configuration-group names (catalog)

LiF / Li-rich

- LiF64_kjpaw, LiF64_kjpaw_v2
- LiF64_kjpaw_NPT_v3, LiF64_kjpaw_final
- LiFinterface_kjpaw_v1, LiFinterface_kjpaw_v2
- LiwithF_NPT_final, LiwithF_isolated
- LiwithBF, LiwithBF_NPT_final

Salts, B–Li, cells, ionics

- LiBF4, LiBF4_NPT_final, LiBF4v4
- LiBF4_relax
- BLi3_interface_NPT, BLi3_isolated
- BCC_54_kjpaw_NPT, BCC54_isolated
- EMIM_TFSI_BF4, EMIMLiTFSI_anode_rv2
- special

The FitSNAP model we have

- **FitSNAP stack** in words:
  - **Scraper** — reads quantum-chemistry frames (e.g. JSON) into structures.
  - **Calculator** — builds **SNAP bispectrum** descriptors per atom in LAMMPS (radial cutoff, `twojmax`, chemical channels, etc.).
  - **Solver** — **PyTorch** MLP.
  - **Layers:** PyTorch MLP `num_desc` $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 1, `multi_element_option = 2` (per-element nets).
  - **Training:** learning rate 5e-6, 70 epochs, batch 50.

Metrics

Errors	Li-F^{full}	Li-B-F^{full}	Li-F^{comp}	Li-B-F^{comp}	Ta[1]	W-BE[2]	Si-Mo-S[3]
E MAE_{Train}	2.78	2.90	41.84	133.0	99.53	116.85	7.44
E MAE_{Test}	2.83	2.93	42.18	131.1	168.44	120.65	6.91
E $RMSE_{Train}$	3.98	4.09	66.99	195.0	316.76	597.15	10.49
E $RMSE_{Test}$	4.06	4.14	67.47	188.5	535.25	611.06	11.55
F MAE_{Train}	0.0104	0.0184	0.2159	0.5405	0.07	0.30	0.13
F MAE_{Test}	0.0106	0.0188	0.2166	0.5435	0.08	0.30	0.11
F $RMSE_{Train}$	0.0146	0.0303	0.4082	0.9052	0.16	0.79	0.20
F $RMSE_{Test}$	0.0151	0.0313	0.4107	0.9148	0.16	0.71	0.18

Figure: Metrics obtained by two models

What the MACE model is

MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields

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Figure: I. Batatia et al. 2022

Message Passing Neural Networks

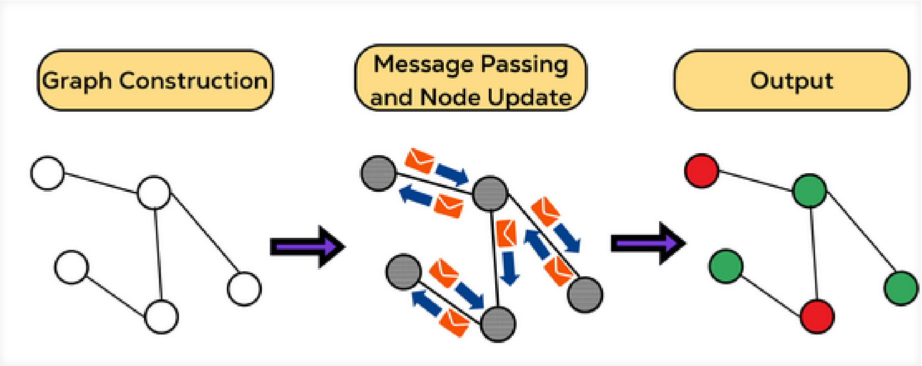


Figure: Message Passing Neural Network Concept

MACE Step (1): Atom Embedding

A molecule or crystal is represented as a **graph**: atoms are nodes, edges connect pairs within a cutoff radius r_c. Each atom i is assigned an initial feature vector:

$$\mathrm{h}_i^{(0)} = \mathrm{W}_{Z_i}$$

a learned embedding indexed by atomic number Z_i (lookup table). **Goal:** produce, for each atom, a scalar energy E_i that reflects its local chemical environment. The total energy is $E = \sum_i E_i$.

Depth on MPNN

- $t = 0$, the atom/node only can see itself.
- $t = 1$, the atom can see its neighborhoods - **TWO BODY**.
- $t = 2$, the atom can see the neighborhoods from its neighborhoods - TRIPLETS - **THREE BODY**

Three and higher order bodies are computationally expensive.

MACE Step (2): Interaction Block

For each atom i, messages from neighbors j are built as:

$$\mathrm{m}_{ij} = \sum_k W_k \, R_k(r_{ij}) \cdot Y_l^m(\hat{r}_{ij}) \otimes \mathrm{h}_j^{(t)}$$

W_k learned matrix.

- $Y_l^m(\hat{r}_{ij})$: **spherical harmonics** — fixed geometric functions encoding the *direction* to neighbor j.
- $R_k(r_{ij})$: **radial network** — a small MLP acting on the scalar distance. *This is where most parameters live.*
- $\otimes$: tensor product combining neighbor features with edge geometry, preserving equivariance via Clebsch–Gordan coefficients.

Messages are aggregated: $\mathrm{m}_i = \sum_{j \in \mathcal{N}(i)} \mathrm{m}_{ij}$

MACE Step (3): Equivariant Product Block

After aggregating neighbors, atom i holds a feature vector m_i encoding its local environment.

The problem: one message passing layer only captures pairwise (i, j) interactions — that is **2-body**.

The MACE idea: take tensor products of m_i with itself ν times:

$$\mathrm{m}_i \otimes \mathrm{m}_i \otimes \cdots \quad (\nu \text{ times})$$

This mixes contributions from *different* neighbors j, k simultaneously, producing $(\nu + 1)$-**body correlations** without any extra message passing.

- With $\nu = 2$: reaches **3-body** after layer 1, compounding across layers.
- The CG coefficients ensure the result stays **equivariant** — no parameters, fixed by symmetry.

MACE Step (4): Readout

After T interaction layers, each atom i holds a feature vector $\mathrm{h}_i^{(t)}$ with components of different equivariance order L. Each layer contributes a partial atomic energy via its $L = 0$ features:

$$E_i = E_i^{(0)} + E_i^{(1)} + \cdots + E_i^{(T)}$$

where each $E_i^{(t)}$ is a linear projection of $\mathrm{h}_i^{(t)}\big|_{L=0}$, except the last layer which uses a small MLP.

The total energy and forces are then:

$$E = \sum_i E_i, \qquad \mathrm{F}_i = -\frac{\partial E}{\partial \mathrm{r}_i}$$

Forces are free — no extra network needed, just automatic differentiation through the entire graph.

Parameters Distribution

Block	Parameters	Share
`interactions` (2 layers)	**3,163,392**	**~82%**
`products` (2 layers, correlation / symmetric contraction)	**670,720**	**~17%**
`node_embedding`	11,392	~0.3%
`readouts`	2,192	~0.1%
Total	3,847,696	100%

Submodule (layer 0)	Params	Role (intuition)
`skip_tp`	**~1.46M**	Main **equivariant tensor-product / mixing** path (the heavy part)
`linear`	~65k	Channel mixing linear on node features
`conv_tp_weights`	**~42k**	**Radial MLP** output → weights for the convolution (your `[64,64,64]` lives here)

MACE-OFF23: Training Dataset

MACE-OFF23 is a *transferable* foundation model for organic chemistry — not trained on specific molecules, but on a broad chemical space.

Dataset: SPICE

- Covers **10 elements**: H, C, N, O, F, P, S, Cl, Br, I
- Only **neutral molecules** (no ion pairs)

- 128 feature channels
- $L = 0$ — **invariant messages only**, no equivariant features beyond scalars
- Faster and cheaper than the full $L = 2$ variant, at some cost in accuracy

Key difference from the NeurIPS paper: those models were specialist (one molecule at a time). MACE-OFF23 is a **general-purpose** force field covering broad organic chemistry.

Zero-Shot MACE-MP Evaluation on $LiBF_4$

Setup: MACE-MP (small, $L = 0$) evaluated zero-shot on $LiBF_4$ — an ionic electrolyte system *outside* the MACE-OFF23 training domain (neutral organics only).

Results: 175 test configurations

Metric	Value	Interpretation
Energy MAE	10,692 eV	reference offset
Energy Std	0.42 eV	relative energies OK
Energy/atom MAE	594 eV/atom	reference offset
Force RMSE	0.44 eV/Å	marginal
Force MAE	0.68 eV/Å	marginal
Max $\lvert\Delta F\rvert$	1.57 eV/Å	largest errors

Zero Shot

Key observations:

- Energy MAE $\gg$ Energy Std $\Rightarrow$ the error is a **systematic constant offset**, not random noise.
- Caused by mismatched **atomic reference energies** $E_0(Z_i)$ between FitSNAP DFT data and MACE-MP training.
- After offset correction, **relative energies are reasonable** (Std $\approx$ 0.42 eV).
- Force errors are moderate — the model captures **some** local chemistry of Li and BF_4^- zero-shot.

Conclusion: promising zero-shot force prediction; energy requires reference alignment before use in MD.

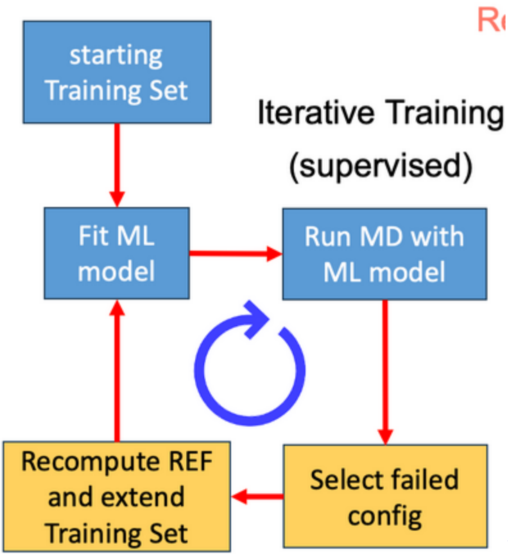
Challenges of Fine-Tuning MACE on a Specific Dataset

- **Reference energy mismatch (E_0 problem)**
  The foundation model and the target DFT dataset use different atomic reference energies. A linear regression over the training set is required to align them before fine-tuning.
- **DFT functional mismatch**
  MACE-MP was trained with very specific data, if different functional (DFT), the model must *unlearn* systematic biases — which can hurt generalization.
- **Out-of-domain chemistry**
  $LiBF_4$ contains ionic species and Li, which are outside the MACE-OFF23 organic training domain. The model has no prior knowledge of ionic interactions, charge transfer, or metal coordination.

Challenges

- **Data scarcity**
  Fine-tuning on a small dataset risks **catastrophic forgetting** — the model overfits to the new data and loses the general knowledge from pretraining.
- **Force/energy weight balance**
  The loss $\mathcal{L} = \lambda_E \mathcal{L}_E + \lambda_F \mathcal{L}_F$ is sensitive to the choice of λ_E, λ_F. Wrong balance leads to models that predict energies well but produce unphysical forces or vice versa.

Iterative Learning



Active Learning